Josh Talks Voice Arena — Speaker Diarization Benchmark

Executive Evaluation & Architecture Comparison across Telephony, Broadcast & Meeting Workflows

Collar: 0.25s | Overlap: Evaluated
Date: October 2026
Corpus: Voice Arena (10 Systems)

CALL RECORDINGS (8 kHz)
BUT-FIT VBx / NeMo Clustering
Score: **80.31** / **79.26** | DER: **68.8%**
Confusion: **2.02%** (vs pyannote 27.8%)

LIVE TRANSCRIPTION (Overlap)
ESPnet EEND-EDA / NeMo Sortformer
Score: **94.14** / **93.11** | DER: **20.5%**
Tracks rapid overlapping speaker turns

OFFICE ROOM MEETINGS
NeMo Streaming Sortformer
Score: **99.83** | DER: **1.12%**
0.0% Miss/Conf | CPU ~7.6s Latency

Category Benchmark Results & Production Leaderboards

Domain Category	Error Weights (Miss/FA/Conf)	Category Champion	Score	DER %	Leading Architecture	Core Production Characteristic
Call Recordings	0.40 / 0.25 / 0.35	Model 10: BUT-FIT VBx	**80.31**	68.80%	Bayesian HMM + x-vector	Temporal smoothing prevents telephony confusion; NeMo Clustering achieves 79.26.
Live Transcription	0.50 / 0.20 / 0.30	Model 7: ESPnet EEND-EDA	**94.14**	20.50%	Encoder-Decoder Attractor	Multi-speaker attention resolves overlap; NeMo Sortformer achieves 93.11.
Office Meetings	0.45 / 0.15 / 0.40	**Model 4: NeMo Streaming**	**99.83**	1.12%	Causal Rolling Sortformer	Zero speaker confusion (0.00%); streaming buffer matches offline accuracy.
General Aggregate	0.40 / 0.30 / 0.30	**Model 2: NeMo Sortformer**	**92.12**	24.31%	Self-Attention Sortformer	Highest cross-domain stability across clean dialogue and degraded audio.

1. Framework & Custom Scoring Rationale

- **Why Category Scores:** Equal-cost DER fails in enterprise setups. Custom Score = 100 × max(0, 1 – CWE) applies domain-specific penalties.
- **Call Recordings (w_conf=0.35):** Confusion ruins QA audits & compliance tracking; missed turns lose statutory disclosures.
- **Live Transcriptions (w_miss=0.50):** Missed speech directly drops words in downstream speech-to-text pipelines.
- **Office Meetings (w_conf=0.40):** Confusion scrambles action-item assignments in automated minutes.

2. Key Empirical Findings Across Systems

- **Streaming Sortformer Offline Parity:** Model 4 matches offline accuracy (99.83 vs 99.75 in meetings; 93.07 vs 93.11 in live audio) on CPU (~7.65s runtime).
- **Clustering Dominates Telephony:** Models 10 & 3 achieve 2.02%–2.10% confusion on 8 kHz calls vs pyannote's 27.81% confusion.
- **VAD Sensitivity Trade-off:** pyannote achieves low call miss (6.30%), but triggers elevated false alarms (63.04% on calls, 13.19% on live audio).

3. Architectural Paradigms Evaluated

- **Self-Attention Sortformers (Models 2, 4):** Transformer sorting predicting multi-speaker masks without heuristic clustering.
- **Modular Neural Clustering (Models 1, 3, 5, 6, 8):** Dedicated VAD + deep embeddings (TitaNet, ECAPA, ResNet, x-vector) + spectral/AHC clustering.
- **End-to-End Multi-Speaker (Model 7):** Attractor networks predicting overlapping speaker tracks natively.
- **Probabilistic State Models (Models 9, 10):** Hidden Markov Models with Variational Bayes or recurrent transitions.

4. Strategic Production Takeaways

- **Real-Time ASR & Subtitling:** Deploy **NeMo Streaming Sortformer (Model 4)** for zero session re-clustering and instant turn emissions.
- **Contact Center QA & Audit:** Deploy **NeMo Clustering Diarizer (Model 3)** or **BUT-FIT VBx (Model 10)** for robust speaker attribution on 8 kHz channels.
- **Multi-Party Overlap:** Deploy **NeMo Sortformer (Model 2)** or **ESPnet (Model 7)** to separate simultaneous overlapping speakers.

Data Quality Notice (call_input02): Ground-truth reference spans 400.35s–479.77s while the physical file is 120.00s (reference duration = 0.0s, yielding 0/0 division). Officially excluded as INVALID_REFERENCE_DURATION to protect scoring integrity. A –400s offset recovers valid alignment, proving models behave predictably.